

Olga Petrovska, SFHEA

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Personal Statement

Academic leader and **Senior Fellow of the HEA** with **10 years of teaching experience**, expertise in **computing education research** and assessment design, and a **strong project management background**. Leading a teaching attainment group at Swansea University, focusing on **AI in Computer Science** and co-leading an international working group on **GenAI Literacy**. Current research focuses on **pedagogical innovation**, exploring GenAI in Higher Education, sustainability, and ethics. Committed to advancing academic excellence.

Employment

- 2022 –  **Head of Final Year** at Computer Science Department, Swansea University, UK
- Under my leadership (Programme Director cover in 2022/23), our Applied Software Engineering programme achieved the best NSS scores in its history: 87% in learning opportunities, 88% in assessment and feedback, 98% in teaching, 100% in academic support
 - Collaborate with external examiners and industry partners to balance academic standards within the programme with practical skills that enhance graduate employability
 - Lead the assessment review process for the whole programme, ensuring rigorous moderation and adherence to academic standards
- 2020 –  **Teaching Fellow** at Computer Science Department, Swansea University, UK
- Develop and coordinate various undergraduate modules
 - Research the applications, implications, and potential of GenAI in Higher Education
 - Coordinate and mentor teaching staff, facilitating their ongoing professional development (e.g. in project supervision, assessments, integration of GenAI into teaching, etc.)
- 2020 – 2025  **Project Manager** and **Technology Consultant** at Ukraïner.net, remote (*part-time*)
- Designed and delivered technical training for Ukraïner's staff and volunteers
 - Successfully managed several multimedia localisation projects
- 2016 – 2020  **Teaching Assistant** at Swansea University, UK (*part-time*)
- Supported teaching across a variety of modules at the Computer Science Department
- 2013 – 2015  **Project Manager / Team Leader** at Wolfestone, Swansea, UK
- Successfully led a team of five project managers in delivering various localisation and translation projects, earning Employee of the Year in 2014

Positions of Responsibility and Academic Environment

- 2026 –  **Fellow** at Google for Education Higher Education Faculty AI Fellowship
-  **Co-leader** of the ITiCSE Working Group on Improving CS Students' GenAI Literacy
- 2025 –  **Leader** of the "Embracing AI in Computer Science" Working Group within Swansea University's CoSTA (Computer Science Teaching Attainment Group)
-  **Program Committee Member** at The UK and Ireland Computing Education Research Conference (UKICER 2026) [RIPPA Track Chair], Human-Centered AI Education & Practice Conference (HCAI-ep'26) [Poster Track Chair], Innovation and Technology in Computer Science Education (ITiCSE 2026), and ACM Virtual Global Computing Education Conference (SIGCSE Virtual 2026), SIGCSE Technical Symposium 2026 and 2027, Koli Calling 2025 and 2026, Computing Education Practice (CEP 2026 and 2027), and Continuity, Computability, Constructivity (CCC 2025)

Positions of Responsibility and Academic Environment (continued)

- 2025 – 2026  **Co-leader** of the ITiCSE Working Group on Refactoring Upper Computing Courses to Leverage GenAI
-  **Reviewer** of HEA fellowship applications under Swansea University's Advance HE-accredited Celebration of Professional Recognition (CoPR) programme
- 2024 –  **Reviewer** for the several international journals (Frontiers in Education, Information and Software Technology, IEEE Transactions on Learning Technologies, Media and Communication, and IEEE Access) and conferences (CHI'26, OpenEd25, and BCSWomen Lovelace Colloquium)
- 2023 –  **Council Member** at the Association of Computability in Europe
-  **Project Support Officer** at the UK-Ukraine Academic Mentorship Scheme
- 2020 –  **Treasurer** at the British Colloquium for Theoretical Computer Science

Education

- 2016 – 2021  **Ph.D., Computer Science**
Swansea University, UK
Thesis: *Enhanced Realizability Interpretation for Program Extraction*
- 2015 – 2016  **M.Sc., Computer Science**
Swansea University, UK
Thesis: *Prawf: Interactive Proof Assistant for Predicate Logic*
- 2006 – 2007  **Master of Philology, English Language and Literature**
Chernivtsi National University, Ukraine
Thesis: *The Functional-Semantic Field of Quantity in John Fowles' Discourse*
- 2002 – 2006  **Bachelor of Pedagogy, Pedagogical Education**
Ternopil National Pedagogical University, Ukraine

Teaching Experience and Course Design

- CS Fundamentals  **Algorithms and Automata:** module design & delivery since 2021
-  **Computational Thinking:** microcredential module delivery in 2022/2023
-  **Discrete Mathematics for Computer Science:** module delivery in 2020/2021
-  **Modelling Computing Systems:** module support since 2021, including delivery of tutorials
- Project Management  **Project Planning and Management:** module design & delivery since 2021
-  **Software Engineering Project Management:** microcredential module delivery in 2022/2023
-  **Software Engineering Project Planning and Management:** module co-delivery in 2022/2023
-  **Project Specification and Development:** module coordination since 2021
-  **Project Implementation and Dissertation:** module coordination since 2021
- Professional Issues  **Research Methods** co-teaching since 2026
-  **Legal, Social, Ethical, and Professional Issues:** module design & delivery since 2026
-  **Professional Issues of Software Engineering:** module design & delivery in 2021 - 2025
- CS Other  **Introduction to Data Science:** module support since 2026

Teaching Experience and Course Design (continued)

	■ Software Testing: microcredential module design and delivery in 2025/2026
	■ Artificial Intelligence: co-design of symbolic AI components in 2024/2025
	■ Human-Computer Interaction: delivery of selected components in 2021/2022
	■ Computer Security: module co-delivery in 2020/2021
Other Provision	■ Designed and delivered a range of short courses and workshops for teachers, college students, and general public on Artificial Intelligence, Cyber Security, Graphic Design, Web Development, Excel and other topics since 2021.
Other	■ Provided support with labs, tutorials and assessment in a variety of CS modules as a Teaching Assistant between 2016 and 2020: Declarative Programming, Critical System / High Integrity Systems, Logic in Computer Science, Automata Theory and Formal Languages, Introduction to Algorithms and Data Structures

CPD Courses and Certifications

2025	■ Personal Development Course for External Examiners , Advance HE
	■ Level 4 NVQ Diploma in Management , ILM
	■ Project Management Professional Certificate , Google
	■ Sustainability in the Digital Age: Energy-Efficient Software Development , openHPI
	■ Sustainability in the Digital Age: Environmental Impacts of AI Systems , openHPI
2024	■ AI in Education: Leveraging ChatGPT for Teaching , University of Pennsylvania
2023	■ Data Science Bootcamp , openHPI
2022	■ Sustainable Software Engineering , openHPI

Publications

Conference Proceedings

- 1 D. J. Bouvier, B. P. Cipriano, R. Glassey, **O. Petrovska**, E. Anderson, A. Birillo, R. Dougherty, R. Pettit, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, and A. Vadaparty, "The Rest of the Robots: Generative AI in Post-introductory Computing Education," in *Proceedings of the 2025 Working Group Reports on Innovation and Technology in Computer Science Education*, ser. ITiCSE-WGR 2025, Netherlands: ACM, 2026, pp. 61–107, ISBN: 9798400721670.  DOI: 10.1145/3760545.3783970.
- 2 A. Davis, A. Holland-Minkley, M. Olsen, A. Brady, D. Anderson, B. M. Anthony, D. G. Cooper, R. Engineer, R. Lamprecht, A. Petersen, **O. Petrovska**, K. R. B. Schmitt, K. Subramanian, and H. W. Walker, "Essential computing capabilities: Identifying foundational underpinnings for computing curricula," in *Proceedings of the 2nd ACM Virtual Global Computing Education Conference V.2 (SIGCSE Virtual 2026)*, In press, New York, NY, USA: ACM, 2026.
- 3 N. Micallef and **O. Petrovska**, "Structuring Transparency: Developing Domain-Specific Generative AI Declaration Frameworks in Higher Education," in *Proceedings of the International Symposium on Artificial Intelligence and the Transformation of Higher Education (AITHE)*, Accepted. Swansea (UK), July 1–3, 2026.  DOI: <https://doi.org/10.48550/arXiv.2606.13389>.
- 4 **O. Petrovska** and C. Hopkins, "Propaganda or Exposure? A Video-Based Group Assessment Exploring Ethics in a Fictional Technological Society," in *Proceedings of the 10th Computing Education Practice*, ser. CEP '26, ACM, 2026, pp. 33–36, ISBN: 9798400721212.  DOI: 10.1145/3772338.3772352.

- 5 **O. Petrovska**, M. Venn-Wycherley, C. Clarkson, and F. Pantekis, “Building foundational ai literacy through modular generative ai workshops,” in *Proceedings of the International Symposium on Artificial Intelligence and the Transformation of Higher Education (AITHE 2026)*, Accepted. Swansea (UK), July 1-3., 2026.  DOI: <https://doi.org/10.13140/RG.2.2.33291.86563>.
- 6 F. Pantekis and **O. Petrovska**, “Pre-recorded ‘Vivas’ for Assessing Code understanding Amidst AI-Assisted Programming,” in *Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education, Volume 2 (ITiCSE 2026)*, In press, New York, NY, USA: ACM, 2026.  DOI: [10.1145/3803401.3811969](https://doi.org/10.1145/3803401.3811969)[not_yet_active].
- 7 B. Pereira Cipriano, **O. Petrovska**, N. Pombo, L. Battestilli, L. Farinetti, R. Glassey, M. Kasinidou, O. Olayinka, A. Shah, A. Steinmaurer, R. Vaidhiyanathan, and W. James, “Towards Improving CS Students’ Generative AI Literacy,” in *Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education V. 2*, ser. ITiCSE 2026, Spain: ACM, 2026, pp. 789–790, ISBN: 9798400726330.  DOI: [10.1145/3803401.3812055](https://doi.org/10.1145/3803401.3812055).
- 8 J. Roberts and **O. Petrovska**, “Authentic Assessment in Discrete Mathematics: Real-World Modelling Using Logic and Set Theory,” in *Proceedings of the 31st ACM Conference on Innovation and Technology in Computer Science Education, Volume 2 (ITiCSE 2026)*, In press, New York, NY, USA: ACM, 2026.  DOI: [10.1145/3803401.3811976](https://doi.org/10.1145/3803401.3811976)[not_yet_active].
- 9 O. Andrei, S. W. Nabi, M. Barr, and **O. Petrovska**, “Integrating Socially Responsible Computing Competencies in Computer Science and Software Engineering Education,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, ACM, Jan. 2025, pp. 36–37, ISBN: 9798400711725.  DOI: [10.1145/3702212.3702225](https://doi.org/10.1145/3702212.3702225).
- 10 D. J. Bouvier, B. P. Cipriano, R. Glassey, R. Pettit, E. Anderson, A. Birillo, R. Dougherty, O. Hazzan, **O. Petrovska**, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, A. Vadaparty, and G. R. Yeluripati, “GenAI Integration in Upper-Level Computing Courses,” in *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2*, ser. ITiCSE 2025, Nijmegen, Netherlands: ACM, Jun. 2025, pp. 691–692, ISBN: 9798400715693.  DOI: [10.1145/3724389.3731276](https://doi.org/10.1145/3724389.3731276).
- 11 L. Clift and **O. Petrovska**, “Learning without Limits: Analysing the Usage of Generative AI in a Summative Assessment,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, ACM, Jan. 2025, pp. 5–8, ISBN: 9798400711725.  DOI: [10.1145/3702212.3702214](https://doi.org/10.1145/3702212.3702214).
- 12 **O. Petrovska**, L. Clift, and F. Pantekis, “Video in Assessments for Soft Skill Development and Evaluation,” in *Proceedings of the 2025 Conference on UK and Ireland Computing Education Research*, ser. UKICER’25, ACM, Sep. 2025, ISBN: 9798400720789.  DOI: [10.1145/3754508.3754531](https://doi.org/10.1145/3754508.3754531).
- 13 **O. Petrovska**, R. Pearsall, and L. Clift, “Assessing Software Engineering Students’ Analytical Skills in the Era of Generative ai,” in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP’25, ACM, Jan. 2025, p. 34, ISBN: 9798400711725.  DOI: [10.1145/3702212.3702223](https://doi.org/10.1145/3702212.3702223).
- 14 **O. Petrovska**, L. Clift, F. Moller, and R. Pearsall, “Incorporating Generative AI into Software Development Education,” in *Proceedings of the 8th Conference on Computing Education Practice*, ser. CEP’24, Durham, United Kingdom: ACM, Jan. 2024, pp. 37–40, ISBN: 9798400709326.  DOI: [10.1145/3633053.3633057](https://doi.org/10.1145/3633053.3633057).
- 15 **O. Petrovska**, L. Clift, and F. Moller, “Generative AI in Software Development Education: Insights from a Degree Apprenticeship programme,” in *Proceedings of the 2023 Conference on United Kingdom & Ireland Computing Education Research*, ser. UKICER’23, Swansea, Wales UK: ACM, Sep. 2023, ISBN: 9798400708763.  DOI: [10.1145/3610969.3611132](https://doi.org/10.1145/3610969.3611132).
- 16 U. Berger, **O. Petrovska**, and H. Tsuiki, “Prawf: An Interactive Proof System for Program Extraction,” in *Beyond the Horizon of Computability*, M. Anselmo, G. Della Vedova, F. Manea, and A. Pauly, Eds., Cham: Springer International Publishing, 2020, pp. 137–148, ISBN: 978-3-030-51466-2.  DOI: [10.1007/978-3-030-51466-2_12](https://doi.org/10.1007/978-3-030-51466-2_12).

- 17 U. Berger and O. Petrovska, “Optimized Program Extraction for Induction and Coinduction,” in *Sailing Routes in the World of Computation*, F. Manea, R. G. Miller, and D. Nowotka, Eds., Cham: Springer International Publishing, 2018, pp. 70–80, ISBN: 978-3-319-94418-0.  DOI: 10.1007/978-3-319-94418-0_7.

Other

- 1 O. Petrovska, C. Clarkson, F. Pantekis, M. Venn-Wycherley, and F. Boy, *Modular Generative AI Training to Promote AI Literacy Across the University*, SUSALT 2025 - Enhancing Our Practice: Share - Learn - Create, Swansea, 2025.  DOI: 10.5281/zenodo.18637047.
- 2 O. Petrovska, *Course Portfolio on Algorithms and Automata (Apprenticeship)*, 2024.  DOI: 10.5281/zenodo.14448266.
- 3 O. Petrovska, *Enhanced Realizability Interpretation for Program Extraction*, 2021.  DOI: 10.23889/suthesis.57831.

Event Organisation

- Jun 2026  **Logic Colloquium 2026** – Local organiser¹
- Sep 2025  **CCC 2025** (Continuity, Computability, Constructivity) – Local organiser
- Mar 2025  **UK-Ukraine Research Twinning Showcase Event 2025** at the Liverpool University – Co-organiser
- Jul 2024  **CCA 2024** (International Conference on Computability and Complexity in Analysis) – Local organiser
- May 2024  **Agda Implementors’ Meeting XXXVIII** – Local organiser
- Sep 2023  **UKICER 2023** (UK and Ireland Computing Education Research) – Local organiser
- Jul 2022  **CiE 2022** (Computability in Europe: Revolutions and Revelations in Computability) – Local organiser
- Apr 2022  **BCTCS 38** (British Colloquium for Theoretical Computer Science) – Co-chair
- Apr 2020  **BCTCS 36** online conference – Local organiser
- Sep 2019  **2nd International Summer School on Proof Theory and Workshop on Proof Theory and its Applications** – Local organiser

Selected Talks

- 1 **A Taste of Technocamps** at the International Workshop on Physical Computing | with Megan Venn-Wycherley (10.08.2026, Uppsala University, Sweden)
- 2 **GenAI for Researchers: Capabilities, Risks, and Responsible Use** at a workshop within the NextGen AI Bootcamp (20.02.2026, Swansea University, Wales)
- 3 **The Use of Video in Assessments for Soft Skill Development and Evaluation** at the University of Nottingham (26.11.2025, online)
- 4 **Smile! You’re Being (Video) Assessed** at the Teaching Hour of the Computer Science Education Group, University of Edinburgh (24.11.2025, online)
- 5 **Modular Generative AI Training to Promote AI Literacy Across the University** at the Swansea University annual Learning and Teaching Conference SUSALT 2025 - Enhancing Our Practice: Share - Learn - Create | with Filippas Pantekis (10.07.2025, Swansea University, Wales)

¹All conferences/events organised at Swansea University, unless stated otherwise.

- 6 **The Art of Teaching Theory Across Diverse Backgrounds** at the British Colloquium for Theoretical Computer Science 2025 (14-16.04.2025, University of Strathclyde, Scotland)
- 7 **GenAI in Education and Research** at the NAIADES (Network for Artificial Intelligence, the Arts, the Digital Economy and Society) Workshop “GenAI: Use and Best Practices” (20.02.2025, Swansea University, Wales)
- 8 **Computer Science Outreach in Higher Education: Challenges and Opportunities** | with Megan Venn-Wycherley (08.01.2025, Durham University, England)
- 9 **Developing Students’ Critical Skills in the AI-enabled World** at the Advance HE Assessment and Feedback Symposium 2024 (05.11.2024, online)
- 10 **How are students using Generative Artificial Intelligence?** (01.11.2024, “Pinch of SALT” podcast)
- 11 **Generative AI and Conversations with Your Students** at the Lunch and Learn Webinar of the Swansea Academy of Learning and Teaching | with Lee Clift (22.10.2024, online)
- 12 **Student vs AI** at the Learning and Teaching Enhancement Centre (LTEC) Conference “GenAI: Challenges and Solutions” (18.09.2024, Swansea University, Wales)
- 13 **Operational Semantics in Prawf** at the Workshop on Proof and Computation 2022 (01.06.2022, Kloster Schlehdorf, Germany)
- 14 **Intuitionistic Fixed Point Logic and Program Extraction** at the Workshop on Continuity, Computability, Constructivity 2018 (24-28.09.2018, University of Algarve, Faro, Portugal)
- 15 **Natural Language Proofs for Program Extraction** at the International Workshop on Proof, Computation, Complexity 2017 (26-27.07.2017, University of Göttingen, Germany)
- 16 **Natural Language Proof Checking in Predicate Logic** at the Workshop on Proofs, Programs, and Verification (19.04.2017, University of Canterbury, Christchurch, New Zealand)

Secondments and Residences

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| Jul-Aug. 2018 | ■ Residence at the Hausdorff Trimester Program (Hausdorff Research Institute for Mathematics, Germany) |
| Mar-Apr. 2018 | ■ Secondment at the Mathematical and Information Sciences Division of the Human and Environmental Studies Faculty (Kyoto University, Japan) |
| Mar-May. 2017 | ■ Secondment at the School of Mathematics and Statistics (University of Canterbury, New Zealand) |

Languages

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| Native or bilingual proficiency | ■ English, Ukrainian, Slovak, Russian |
| Limited working proficiency | ■ German, Polish |
| Elementary proficiency | ■ Welsh, Mandarin Chinese |